



# LOST BOYS CYBER

THREAT INTELLIGENCE • MALWARE ANALYSIS • DETECTION ENGINEERING



## APT28, an evolution of tradecraft

### Threat Actor Profile

|                       |                      |
|-----------------------|----------------------|
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TLP:CLEAR | Lost Boys Cyber | APT28, an evolution of tradecraft - Threat Actor Profile

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# APT28, an evolution of tradecraft

## 1. Executive Summary

APT28, also tracked as Fancy Bear, Forest Blizzard, Sofacy, Pawn Storm, and Sednit, remains one of the most persistent Russian state-sponsored intrusion sets publicly attributed to the GRU's Unit 26165. Sekoia's 11 June 2026 reporting frames the actor's last two decades as an evolution from a stable in-house implant era into a more flexible operating model that combines disposable modules, compromised edge infrastructure, cloud-hosted command channels, and experimental LLM-enabled malware.

The defensive significance is not that APT28 has abandoned its legacy playbook. It is that the group has learned to preserve successful tradecraft while changing the operational surface that defenders monitor. Mature defenders should expect familiar outcomes — credential theft, mailbox compromise, document collection, persistent access, and espionage against government, defense, Ukraine-aligned, and NATO-linked targets — to arrive through less traditional infrastructure such as compromised Ubiquiti EdgeRouters, MikroTik/TP-Link routers, commercial cloud storage, webmail exploitation, and modular tooling with reduced static reuse.

Lost Boys Cyber assesses that organizations with Ukrainian government, defense, civil society, diplomatic, logistics, technology, or NATO support exposure should treat this reporting as a priority detection-engineering and hunt-cycle input. The most actionable defensive work is not a single IOC blocklist; it is telemetry coverage for router abuse, identity attacks, suspicious OAuth/session theft, webmail exploitation, steganographic/cloud C2 patterns, and AI-service contact from endpoints that do not have a business need for runtime LLM API access.

| Field                    | Assessment                                                                                                                                   |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Actor                    | APT28 / Fancy Bear / Forest Blizzard / Sofacy / Pawn Storm / Sednit                                                                          |
| Attribution              | Russia GRU Unit 26165; high confidence from long-running public government and vendor reporting                                              |
| Current tradecraft theme | Stable mission, more distributed and replaceable infrastructure/tooling                                                                      |
| Primary risk             | Espionage, credential theft, mailbox access, data collection, and pre-positioning against Ukraine/NATO-adjacent targets                      |
| Defensive priority       | Hunt identity, edge-infrastructure, cloud-C2, webmail, and LLM-service abuse patterns rather than relying only on malware hashes             |
| Confidence               | High that APT28 is active and evolving; medium-to-high on individual newer campaign linkages where public reporting uses moderate confidence |

## 2. Intelligence Requirement

This report answers the priority question: how has APT28's tradecraft changed, and what should defenders adjust in detection, hunting, and hardening programs as a result?

| Intelligence Requirement                          | Current Answer                                                                                                                                                          | Confidence | Defensive Implication                                                                                               |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|---------------------------------------------------------------------------------------------------------------------|
| Has APT28 materially changed its operating model? | Yes. Public reporting describes a shift from long-lived implant chains toward modular tooling, compromised edge infrastructure, cloud-based C2, and LLM-enabled command | High       | Detection programs must focus on behaviors and infrastructure classes, not only historic Sofacy/X-Agent signatures. |

|                                              |                                                                                                                                                                              |             |                                                                                                                                                            |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                              | generation experiments.                                                                                                                                                      |             |                                                                                                                                                            |
| Which environments are most exposed?         | Ukraine, NATO-aligned government, defense, diplomatic, logistics, civil society, technology, and organizations supporting Ukraine remain priority exposure groups.           | High        | Prioritize identity telemetry, mail security, endpoint command telemetry, and edge-device inventory for these business units.                              |
| Are there hard IOCs sufficient for blocking? | The source theme is tradecraft evolution rather than a single malware sample. Hard IOCs are likely campaign-specific and incomplete.                                         | Medium      | Use IOCs as enrichment, but treat analytic coverage and hunt hypotheses as the primary deliverable.                                                        |
| What is new for 2025-2026?                   | Sekoia highlights Operation Phantom Net Voxel, edge-router operations, industrialized Ukrainian targeting, renewed custom implant activity, and LameHug/LLM-enabled malware. | Medium-High | Add detections for commercial cloud storage abuse, steganographic payload retrieval, Hugging Face/LLM API contact, and suspicious router-adjacent traffic. |
| What should be done first?                   | Close identity and edge visibility gaps, then run focused hunts for webmail exploitation, router proxying, OAuth/session theft, and cloud-C2 staging.                        | High        | Assign work to identity, network, endpoint, and messaging owners with measurable telemetry outcomes.                                                       |

### 3. Source Review and Confidence

The primary intake source is Sekoia.io's "APT28, an evolution of tradecraft," published 11 June 2026. The source consolidates Sekoia TDR tracking and references public reporting on APT28's historical and current operations. It aligns with broader government and vendor attribution that ties APT28/Forest Blizzard/Fancy Bear/Sofacy to Russia's GRU Unit 26165.

| Source                                         | Contribution                                                                                                                                 | Reliability | Information Credibility | Analyst Confidence                                                                               |
|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------------------------|--------------------------------------------------------------------------------------------------|
| Sekoia.io, "APT28, an evolution of tradecraft" | Primary synthesis of APT28 tradecraft changes across 2004-2026, including edge infrastructure, Phantom Net Voxel, and LameHug themes.        | B           | 2                       | High for strategic trend; medium for campaign-specific details not independently validated here. |
| Sekoia.io, "APT28 Operation Phantom Net Voxel" | Corroborates weaponized Office lures, COM-hijack DLL persistence, PNG steganography, Covenant Grunt, Koofr, BeardShell, and Icedrive themes. | B           | 2                       | Medium-High.                                                                                     |
| CERT-UA / Ukrainian government reporting       | Corroborates Ukrainian targeting, LameHug                                                                                                    | A/B         | 2                       | Medium-High; attribution caveats                                                                 |

|                                                       |                                                                                   |          |     |                                                                               |
|-------------------------------------------------------|-----------------------------------------------------------------------------------|----------|-----|-------------------------------------------------------------------------------|
| referenced in public summaries                        | disclosure, and campaign focus.                                                   |          |     | remain for some tooling.                                                      |
| MITRE ATT&CK APT28 and LAMEHUG entries                | Provides technique normalization for ATT&CK mapping.                              | A        | 2   | High for technique taxonomy; not a substitute for incident-specific evidence. |
| Prior Lost Boys Cyber APT28 / Forest Blizzard profile | Provides internal baseline on aliases, GRU attribution, and defensive priorities. | Internal | N/A | High for continuity; updated by this report's 2026 tradecraft focus.          |

#### Assumptions and limitations:

- The report treats APT28/Forest Blizzard/Fancy Bear/Sofacy as the same broad GRU Unit 26165-aligned intrusion set, while acknowledging that individual campaigns may carry vendor-specific confidence levels.
- The intake item is actor/tradecraft focused, not sample-backed malware analysis. No malware sample was provided, so detections are behavior and hypothesis oriented.
- Some 2025-2026 campaign details are based on public summaries and Sekoia descriptions rather than direct artifact collection in this task.

## 4. Key Findings

| # | Finding                                                                                                     | Evidence Basis                                                                                                                                                   | Confidence  | Defensive Priority |
|---|-------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------------------|
| 1 | APT28 has moved from a primarily signature-implant defensive problem to a multi-surface tradecraft problem. | Sekoia describes two decades from X-Agent/X-Tunnel and hack-and-leak operations into disposable modules, edge routers, cloud C2, and LLM-enabled malware.        | High        | High               |
| 2 | Compromised edge infrastructure is now a meaningful part of the actor's resilience model.                   | Public summaries cite MooBot-controlled Ubiquiti EdgeRouters, MikroTik/TP-Link activity, traffic proxying, NTLMv2 collection, and phishing landing page hosting. | Medium-High | High               |
| 3 | Operation Phantom Net Voxel indicates renewed custom implant and cloud-C2 investment.                       | Sekoia summaries describe weaponized Office lures, COM hijacking, PNG steganography, Covenant Grunt via Koofr, BeardShell, and Icedrive.                         | Medium-High | High               |
| 4 | LameHug shows experimentation with LLM-mediated operational logic.                                          | Public reporting describes Python-based infostealer behavior, Hugging Face/Qwen model usage, and                                                                 | Medium      | Medium-High        |

|   |                                                        |                                                                                                                                                                                |      |          |
|---|--------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|----------|
|   |                                                        | command generation at runtime; attribution is often moderate confidence.                                                                                                       |      |          |
| 5 | Identity and mailbox access remain central objectives. | APT28 historical operations and recent public reporting emphasize credential theft, OAuth/session theft, webmail exploitation, Net-NTLMv2 collection, and document harvesting. | High | Critical |
| 6 | IOC-only defense will age quickly.                     | The observed model uses replaceable modules, compromised third-party infrastructure, and legitimate cloud/AI services that can change without altering mission.                | High | High     |

The combined finding is that APT28 is not simply "using new malware." It is distributing operational risk across identities, routers, cloud providers, mail platforms, and legitimate web services. Defensive teams should therefore build detection around the seams between those systems: impossible or unusual sign-in patterns, unexpected LLM/cloud access, endpoint-to-cloud staging, mail rule abuse, router-adjacent proxy behavior, and Office/webmail exploitation leading to script execution or credential capture.

### MITRE ATT&CK Mapping

| Tactic         | Technique                                     | ID        | Relevance                                                                                         |
|----------------|-----------------------------------------------|-----------|---------------------------------------------------------------------------------------------------|
| Reconnaissance | Phishing for Information                      | T1598     | Credential and mailbox-oriented operations depend on targeted collection and lure preparation.    |
| Initial Access | Phishing: Spearphishing Attachment            | T1566.001 | Weaponized Office lure themes remain visible in Phantom Net Voxel and related reporting.          |
| Initial Access | Exploit Public-Facing Application             | T1190     | Webmail exploitation and vulnerable exposed services provide non-phishing access paths.           |
| Execution      | Command and Scripting Interpreter: PowerShell | T1059.001 | Common for post-compromise triage and collection; use behaviorally where artifacts are not fixed. |
| Execution      | Command and Scripting Interpreter: Python     | T1059.006 | Relevant to LameHug-style Python tooling and lightweight scraping/automation.                     |
| Persistence    | Event Triggered Execution:                    | T1546.015 | Public Phantom Net Voxel                                                                          |

|                     |                                                        |               |                                                                                                            |
|---------------------|--------------------------------------------------------|---------------|------------------------------------------------------------------------------------------------------------|
|                     | Component Object Model Hijacking                       |               | summaries describe COM-hijack DLL persistence.                                                             |
| Defense Evasion     | Masquerading                                           | T1036         | Disposable modules and lures can masquerade as documents, updates, or administrative artifacts.            |
| Defense Evasion     | Obfuscated Files or Information: Steganography         | T1027.003     | PNG steganographic payload retrieval is associated with Phantom Net Voxel reporting.                       |
| Credential Access   | OS Credential Dumping / NTLM Relay-Adjacent Collection | T1003 / T1557 | Edge-router operations and Outlook/NTLMv2 collection themes increase credential exposure.                  |
| Credential Access   | Steal Web Session Cookie                               | T1539         | Mailbox/session persistence and OAuth theft themes are relevant to cloud identity monitoring.              |
| Discovery           | System Information Discovery                           | T1082         | LLM-generated or script-generated commands commonly enumerate host/user context before collection.         |
| Collection          | Data from Local System                                 | T1005         | LameHug and related infostealer descriptions emphasize local Office/PDF/TXT document harvesting.           |
| Command and Control | Web Service                                            | T1102         | Koofr, Icedrive, Hugging Face, and other legitimate services can blend C2/staging with normal web traffic. |
| Exfiltration        | Exfiltration Over Web Service                          | T1567         | Cloud storage and web-service abuse support staged collection and transfer.                                |

## 5. Threat Context

APT28's historical identity is well established: a Russian GRU-linked intrusion set active since at least the mid-2000s, publicly associated with high-profile espionage and influence operations, and tracked across vendors as Fancy Bear, Forest Blizzard, Sofacy, Pawn Storm, and Sednit. Earlier defensive models often centered on custom implant families such as X-Agent and X-Tunnel, phishing infrastructure, and recurring malware/tooling signatures.

Sekoia's 2026 framing is useful because it shows how that model has changed without changing the actor's mission. The actor still pursues strategic access and collection, but the operational stack is now more fragmented. Smaller single-purpose modules, compromised residential or small-business routers, commercial cloud services, and third-party platforms create a higher-friction detection problem for organizations that depend on static malware controls or traditional C2 deny lists.

| Period    | Tradecraft Theme                                 | Representative Activity                                 | Defensive Lesson                                        |
|-----------|--------------------------------------------------|---------------------------------------------------------|---------------------------------------------------------|
| 2004-2018 | Signature implant era and hack-and-leak playbook | Stable in-house implant chains such as X-Agent/X-Tunnel | Preserve historical detection, but do not assume reused |

|           |                                              |                                                                                                           |                                                                                                                |
|-----------|----------------------------------------------|-----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
|           |                                              | across high-profile breaches.                                                                             | implants will be present.                                                                                      |
| 2022-2024 | Disposable, single-task modules              | More modular operations with less durable malware reuse.                                                  | Detect execution patterns, staging behavior, and identity abuse rather than waiting for known family names.    |
| 2023-2026 | Operational infrastructure moves to the edge | MooBot/EdgeRouter abuse, router proxying, phishing hosting, NTLMv2 collection, FrostArmada router themes. | Inventory and monitor edge devices; treat outbound/residential proxy anomalies as threat signals.              |
| 2023-2025 | Industrialized Ukrainian target collection   | Ukraine government/civil society/defense-aligned targeting and collection against documents and mail.     | Raise priority for Ukrainian and NATO support business units, partners, and mail tenants.                      |
| 2024-2026 | Renewed custom implant and cloud C2 activity | Phantom Net Voxel, steganography, COM hijacking, Covenant Grunt, Koofr/Icedrive.                          | Monitor cloud-storage access from endpoints and unusual payload retrieval from image/document workflows.       |
| 2025-2026 | LLM-mediated malware experimentation         | LameHug-style runtime command generation via legitimate AI services.                                      | Baseline approved AI-service use and alert on endpoint processes invoking LLM APIs outside expected workflows. |

### Defensive implications by environment

| Environment                      | Exposure                                                                          | What to validate now                                                                                                                            |
|----------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Microsoft 365 / Google Workspace | Mailbox compromise, OAuth/session theft, phishing, mail rule abuse.               | Audit sign-in risk, impossible travel, device compliance, OAuth app grants, inbox rules, legacy auth, and external forwarding.                  |
| Endpoint fleet                   | Office lure execution, script execution, local document harvesting, cloud upload. | Ensure EDR captures process command line, parent-child relationships, script block logging, archive extraction, and cloud endpoint connections. |
| Edge routers / VPN / firewalls   | Proxying, phishing hosting, credential capture, persistence outside EDR coverage. | Inventory internet-facing devices, firmware, exposed admin panels, DNS/HTTP proxy behavior, and unexplained outbound sessions.                  |
| Webmail / collaboration systems  | XSS/exploitation, credential harvesting, session theft.                           | Patch Roundcube/Zimbra/MDaemon/Horde-like platforms; monitor new rules, abnormal user agents, and unusual token use.                            |
| Cloud and SaaS services          | C2/staging through legitimate platforms.                                          | Create allowlists for business-approved Koofr/Icedrive/Hugging Face/AI-service use; alert on endpoint-to-service spikes.                        |

## 6. Detection and Hunting

Detection should prioritize behavior clusters that survive infrastructure churn. The queries below are starting points and must be tuned to local table names, approved cloud-service lists, and normal AI-service use.

## 6.1 Detection Coverage Matrix

| Signal                                                         | Telemetry Source                          | Analytic Goal                                                          | Priority    |
|----------------------------------------------------------------|-------------------------------------------|------------------------------------------------------------------------|-------------|
| Office or archive lure spawning script interpreters            | EDR process telemetry                     | Catch spearphishing execution chains and document-borne payloads.      | High        |
| Endpoint contact to unsanctioned cloud storage or AI/LLM APIs  | Proxy, DNS, EDR network events            | Surface cloud-C2, staging, or LameHug-like LLM command generation.     | High        |
| PNG/image download followed by DLL/script execution            | EDR file + network + process              | Detect steganographic payload staging and execution.                   | Medium-High |
| COM hijack registry modification and DLL load                  | Windows registry and image-load telemetry | Identify Phantom Net Voxel-like persistence.                           | High        |
| Suspicious OAuth consent, mailbox forwarding, or inbox rules   | M365/Google audit logs                    | Catch credential/session takeover and mailbox persistence.             | Critical    |
| Traffic relayed through residential/edge-router infrastructure | Firewall, proxy, VPN, DNS                 | Identify proxying through compromised routers or unusual egress paths. | Medium-High |
| Net-NTLMv2 collection indicators                               | Windows auth, SMB, WebDAV, proxy logs     | Detect credential coercion or hash collection activity.                | High        |

## 6.2 Sigma Detection: Office Lure to Script or Cloud Staging

```

title: APT28-Style Office Lure Spawns Script Interpreter or Cloud Staging Tool
id: 91bb3e9a-apt28-office-cloud-staging
status: experimental
description: Detects Office or document-reader processes spawning script interpreters, curl-like
transfer tools, or archive utilities consistent with spearphishing execution chains used by APT28-style
operations.
author: Lost Boys Cyber
date: 2026/06/17
references:
  - https://blog.sekoia.io/apt28-an-evolution-of-tradecraft/
logsource:
  category: process_creation
  product: windows
detection:
  parent_office:
    ParentImage|endswith:
      - '\\\\WINWORD.EXE'
      - '\\\\EXCEL.EXE'
      - '\\\\POWERPNT.EXE'
      - '\\\\OUTLOOK.EXE'
      - '\\\\AcroRd32.exe'
      - '\\\\Acrobat.exe'
  child_suspicious:
    Image|endswith:
      - '\\\\powershell.exe'
      - '\\\\pwsh.exe'
      - '\\\\cmd.exe'
      - '\\\\wscript.exe'
      - '\\\\cscript.exe'
      - '\\\\mshta.exe'
      - '\\\\rundll32.exe'

```

```

- '\\regsvr32.exe'
- '\\python.exe'
- '\\curl.exe'
- '\\bitsadmin.exe'
suspicious_args:
  CommandLine|contains:
    - 'koofr'
    - 'icedrive'
    - 'huggingface.co'
    - 'api-inference.huggingface.co'
    - 'FromBase64String'
    - 'DownloadString'
    - 'Invoke-WebRequest'
    - 'Start-BitsTransfer'
  condition: parent_office and (child_suspicious or suspicious_args)
fields:
- UtcTime
- Computer
- User
- ParentImage
- Image
- CommandLine
falsepositives:
- Enterprise macro workflows, document automation, or sanctioned scripting launched from Office.
level: high

```

### 6.3 KQL Hunt: Cloud, AI, and Steganography-C2 Patterns

```

let lookback = 30d;
let CloudOrAI = dynamic([
  "koofr", "icedrive", "huggingface.co", "api-inference.huggingface.co",
  "hf.co", "qwen", "openrouter", "openai.com", "anthropic.com"
]);
DeviceNetworkEvents
| where Timestamp > ago(lookback)
| where RemoteUrl has_any (CloudOrAI) or InitiatingProcessCommandLine has_any (CloudOrAI)
| where InitiatingProcessFileName !in~ ("msedge.exe", "chrome.exe", "firefox.exe", "Teams.exe",
"Code.exe")
| project Timestamp, DeviceName, InitiatingProcessAccountName, InitiatingProcessFileName,
InitiatingProcessCommandLine, RemoteUrl, RemoteIP, RemotePort, ActionType
| join kind=leftouter (
  DeviceProcessEvents
  | where Timestamp > ago(lookback)
  | where FileName in~ ("powershell.exe", "pwsh.exe", "python.exe", "rundll32.exe", "regsvr32.exe",
"mshta.exe", "wscript.exe", "cscript.exe", "cmd.exe")
  | project DeviceName, ProcessTime=Timestamp, FileName, ProcessCommandLine,
InitiatingProcessParentFileName
) on DeviceName
| where isnull(ProcessTime) or abs(datetime_diff('minute', Timestamp, ProcessTime)) < 60
| order by Timestamp desc

```

### 6.4 KQL Hunt: COM Hijack Persistence Followed by DLL Execution

```

let lookback = 30d;
let ComPaths = dynamic(["\\Software\\Classes\\CLSID\\", "\\Software\\Classes\\WOW6432Node\\CLSID\\",
"InprocServer32"]);
DeviceRegistryEvents
| where Timestamp > ago(lookback)
| where RegistryKey has_any (ComPaths)
| where ActionType in~ ("RegistryValueSet", "RegistryKeyCreated")
| project RegTime=Timestamp, DeviceName, InitiatingProcessAccountName, RegistryKey, RegistryValueName,
RegistryValueData, InitiatingProcessFileName, InitiatingProcessCommandLine
| join kind=leftouter (
  DeviceImageLoadEvents
  | where Timestamp > ago(lookback)
  | where FolderPath endswith ".dll"
  | project LoadTime=Timestamp, DeviceName, FolderPath, SHA256, InitiatingProcessFileName
) on DeviceName

```

```

) on DeviceName
| where isnull(LoadTime) or abs(datetime_diff('hour', RegTime, LoadTime)) < 24
| order by RegTime desc

```

## 6.5 KQL Hunt: Mailbox Persistence and Identity Abuse

```

let lookback = 30d;
let SuspiciousOps = dynamic(["New-InboxRule", "Set-InboxRule", "Set-Mailbox", "Add-MailboxPermission",
"Add-MailboxFolderPermission", "Consent to application", "Add service principal"]);
OfficeActivity
| where TimeGenerated > ago(lookback)
| where Operation has_any (SuspiciousOps)
    or Parameters has_any ("ForwardTo", "RedirectTo", "DeleteMessage", "MarkAsRead", "OAuth",
"offline_access")
| project TimeGenerated, UserId, ClientIP, Operation, Workload, ResultStatus, UserAgent, Parameters
| join kind=leftouter (
    SigninLogs
    | where TimeGenerated > ago(lookback)
    | where RiskLevelAggregated in~ ("medium", "high") or ConditionalAccessStatus =~ "failure"
    | project SigninTime=TimeGenerated, UserPrincipalName, IPAddress, AppDisplayName, UserAgent,
RiskLevelAggregated, ConditionalAccessStatus
) on $left.UserId == $right.UserPrincipalName
| order by TimeGenerated desc

```

## 6.6 SPL Hunt: Router Proxy and NTLM Exposure Indicators

```

index=proxy OR index=firewall OR index=zeek earliest=-30d
((dest_port=445 OR dest_port=139 OR url="*wpad*" OR uri_path="*/autodiscover*" OR
http_user_agent="*Microsoft Office*")
OR (http_user_agent="*python*" OR http_user_agent="*curl*" OR http_user_agent="*Go-http-client*"))
| eval edge_hint=if(match(src_category,"(?:)residential|router|dynamic|unknown") OR
match(src_ip,"^(100\.64\.|10\.|172\.(1[6-9]|2[0-9]|3[0-1])\.|
192\.168\.)"),"possible_edge_or_proxy","internet")
| stats count values(dest_ip) as dest_ip values(url) as url values(http_user_agent) as user_agent by
src_ip, user, edge_hint
| where count > 10 OR edge_hint="possible_edge_or_proxy"
| sort - count

```

## 6.7 YARA-Lite Analytic: LameHug-Style Python LLM Command Generation

```

rule LBC_APT28_LameHug_Style_Python_LLM_Command_Generation
{
    meta:
        author = "Lost Boys Cyber"
        date = "2026-06-17"
        description = "Hunts for Python malware patterns consistent with LLM-mediated command generation
and document harvesting. Use as triage logic, not attribution proof."
        reference = "https://blog.sekoia.io/apt28-an-evolution-of-tradecraft/"
    strings:
        $hf1 = "api-inference.huggingface.co" ascii nocase
        $hf2 = "huggingface" ascii nocase
        $qwen = "Qwen" ascii nocase
        $prompt1 = "generate" ascii nocase
        $prompt2 = "command" ascii nocase
        $doc1 = ".docx" ascii nocase
        $doc2 = ".xlsx" ascii nocase
        $doc3 = ".pdf" ascii nocase
        $py1 = "requests.post" ascii
        $py2 = "subprocess" ascii
        $py3 = "os.walk" ascii
    condition:
        filesize < 5MB and (2 of ($hf*) or $qwen) and 2 of ($prompt*) and 2 of ($doc*) and 1 of ($py*)
}

```

## 6.8 Hunt Workflow

| Step | Action | Expected Output |
|------|--------|-----------------|
|------|--------|-----------------|

|   |                                                                                                                                               |                                                                       |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| 1 | Scope high-risk users and business units tied to Ukraine, defense, government, diplomacy, logistics, and managed-service access.              | Prioritized identity and endpoint list.                               |
| 2 | Run mailbox and OAuth hunts for abnormal forwarding, consent, token, and inbox rule activity.                                                 | Suspicious accounts for containment or deeper review.                 |
| 3 | Run endpoint network hunts for Koofr/Icedrive/Hugging Face/LLM API access from non-browser processes.                                         | Hosts with possible cloud-C2 or LLM-mediated tooling.                 |
| 4 | Review router/VPN/firewall logs for unexplained proxy behavior, unusual admin access, and credential-coercion patterns.                       | Edge devices requiring firmware review, rebuild, or credential reset. |
| 5 | Correlate Office lure execution, steganographic image retrieval, COM hijack registry writes, and DLL loads.                                   | Candidate Phantom Net Voxel-like execution chains.                    |
| 6 | If confirmed, preserve mail audit logs, endpoint triage packages, router configs, DNS/proxy logs, and cloud audit records before remediation. | Evidence package for incident response and future detection tuning.   |

## 7. Recommendations

| Priority | Owner                    | Recommendation                                                                                                                                             | Rationale                                                                                      |
|----------|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Critical | Identity / Messaging     | Enforce phishing-resistant MFA for privileged, executive, and Ukraine/NATO-support users; audit OAuth grants and inbox rules weekly.                       | APT28 operations frequently converge on credential, session, and mailbox access.               |
| Critical | Network / Infrastructure | Inventory internet-facing routers, VPNs, and firewalls; patch firmware; disable exposed admin panels; rotate credentials; monitor outbound proxy behavior. | Edge infrastructure abuse gives the actor stealth and resilience outside EDR coverage.         |
| High     | Detection Engineering    | Deploy and tune the included Office-to-script, COM hijack, cloud/AI-service, and mailbox-persistence analytics.                                            | Behavior coverage survives disposable tooling and infrastructure churn better than IOC blocks. |
| High     | Endpoint Security        | Confirm EDR captures command line, script block, image load, registry, archive extraction, and network telemetry.                                          | Phantom Net Voxel and LamHug-style activity require multi-event correlation.                   |
| High     | Mail / Collaboration     | Patch and harden Roundcube, Zimbra, MDaemon, Horde-like webmail, and collaboration platforms; monitor XSS/exploit                                          | Sekoia reporting includes webmail-focused tradecraft and credential/session collection         |

|             |                  |                                                                                                                                   |                                                                                          |
|-------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
|             |                  | indicators and unusual user agents.                                                                                               | themes.                                                                                  |
| Medium-High | Cloud Governance | Establish approved-use lists for Koofr, Icedrive, Hugging Face, and other AI/cloud services; alert on unapproved endpoint access. | Legitimate web services can be used for C2, staging, or runtime command generation.      |
| Medium      | Threat Intel     | Track campaign-specific IOCs from Sekoia, CERT-UA, Microsoft, ESET, Lumen/Black Lotus Labs, and MITRE, but tag them as ephemeral. | IOCs are useful for scoping but insufficient for this actor's modern tradecraft.         |
| Medium      | IR / Legal       | Pre-stage evidence preservation procedures for mailbox, OAuth, router, and SaaS audit logs.                                       | Edge and cloud evidence can roll quickly and may not be captured by endpoint collection. |

Implementation sequence for a 30-day defensive sprint:

- Week 1: Patch and inventory exposed edge/webmail systems; freeze current identity and mailbox audit retention; identify high-risk users.
- Week 2: Deploy identity/mailbox hunts and review privileged OAuth grants, forwarding rules, and risky sign-ins.
- Week 3: Deploy endpoint/cloud analytics for Office-to-script, COM hijack, cloud-storage, and AI-service access.
- Week 4: Tune detections against false positives, run a cross-domain tabletop around APT28-style mailbox plus edge-router proxy activity, and convert validated hunts into scheduled analytics.

## 8. References

- [1] Sekoia.io, "APT28, an evolution of tradecraft," published 2026-06-11. <https://blog.sekoia.io/apt28-an-evolution-of-tradecraft/>
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